



Sri Lanka Institute of Information Technology

Data Warehousing and Business Intelligence

[IT3021]

Assignment -2

2026

IT23273108 – Y3S2(WE 1.1)

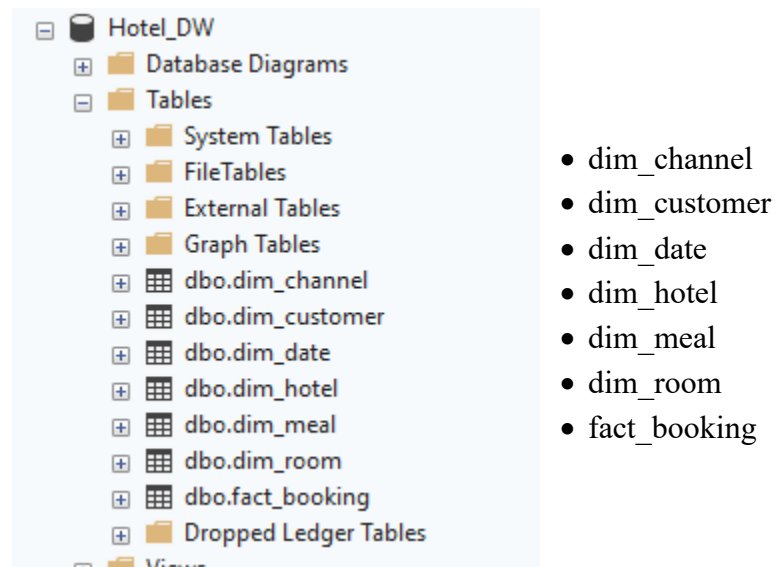
PREMACHANDRA B.M.P.P.

Step 1: Data Source for Assignment 2

Overview

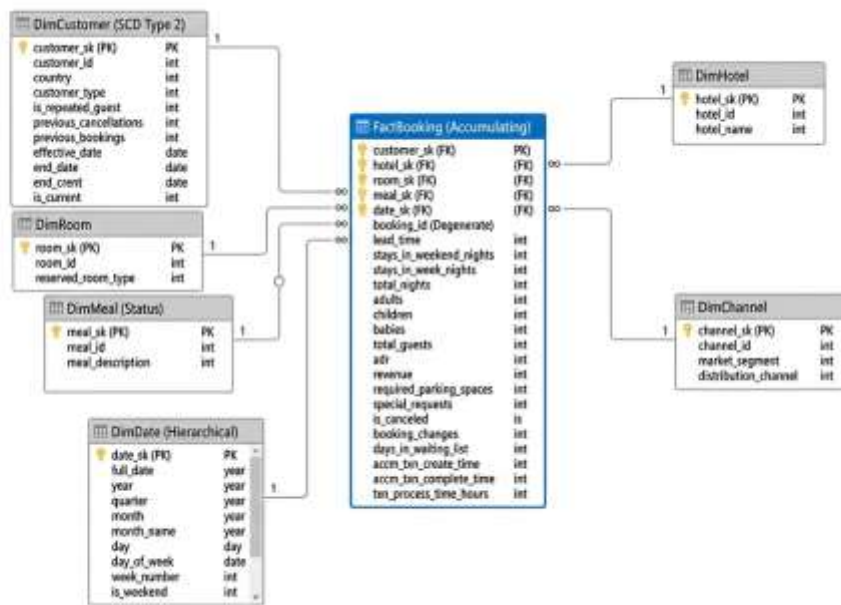
The data warehouse called "**Hotel_DW**" that was built in Assignment 1 is the source of data for this assignment. Microsoft SQL Server 2022 is used as the database management system. The data warehouse is designed for multidimensional analytical processing (OLAP).

My data warehouse contains 6 dimension tables and 1 fact table.



Star Schema Design

The data warehouse follows a **Star Schema** design with one fact table connected to multiple dimension tables. This structure optimizes query performance and enables efficient data aggregation.



Fact Table: fact_booking

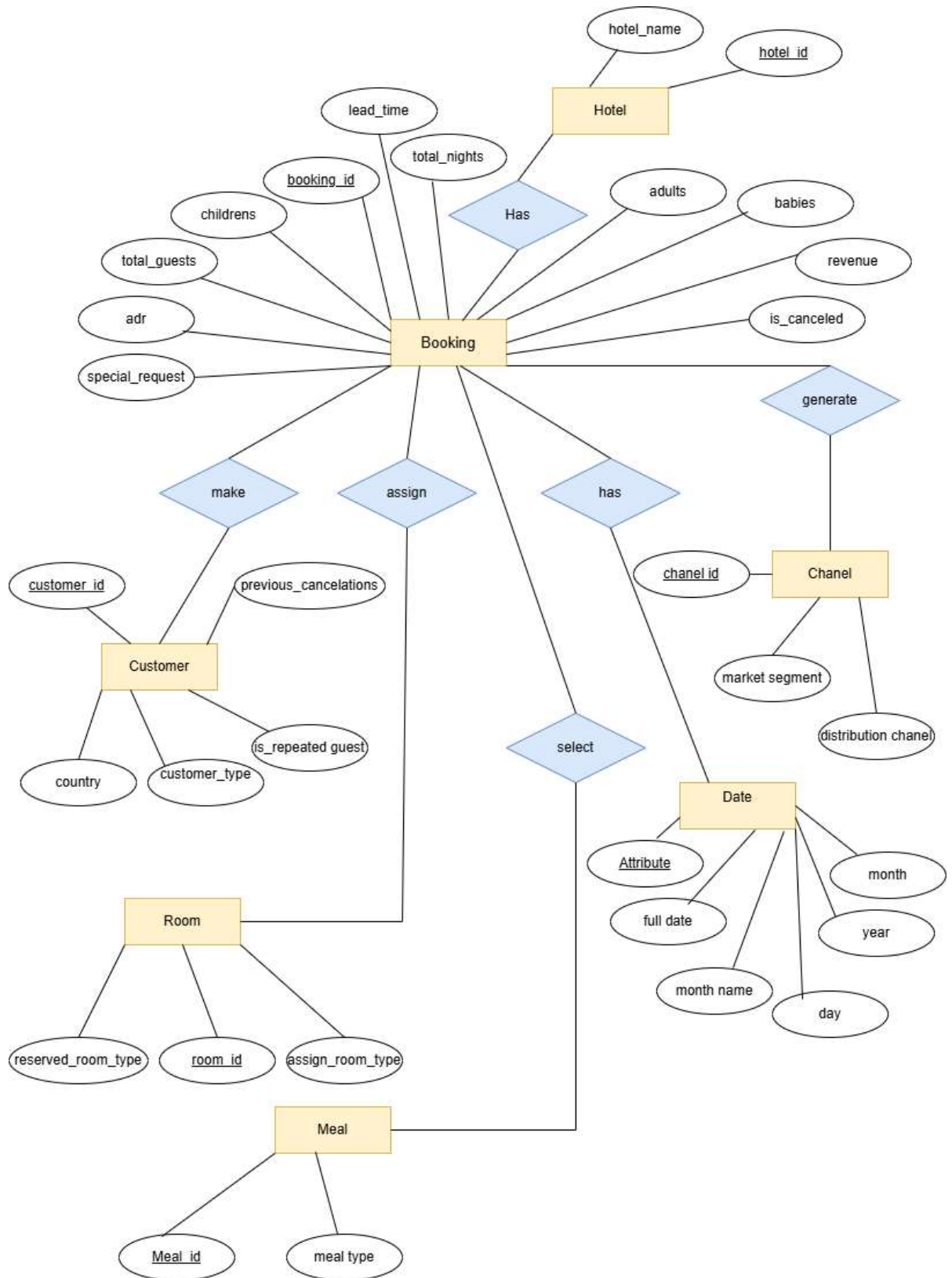
The fact table tracks hotel reservations and stays. It contains:

Type	Description	Example
Measures	Numerical values for analysis	Revenue, ADR, Lead Time, Total Nights
Foreign Keys	Connectors to dimension tables	customer_sk, hotel_sk, room_sk, meal_sk, channel sk, date sk

Dimension Tables

Type	Purpose	Key Attributes
dim_customer	Tracks who booked	customer_type, country, previous_cancellations
dim_hotel	Tracks where they stayed	hotel name
dim_room	Tracks room type	reserved_room_type, assigned_room_type
dim_meal	Tracks meal plan	meal type
dim_channel	Tracks booking source	market_segment, distribution_channel
dim_date	Tracks when	year, quarter, month, day

ER Diagram:



Step 2: SSAS Cube Implementation

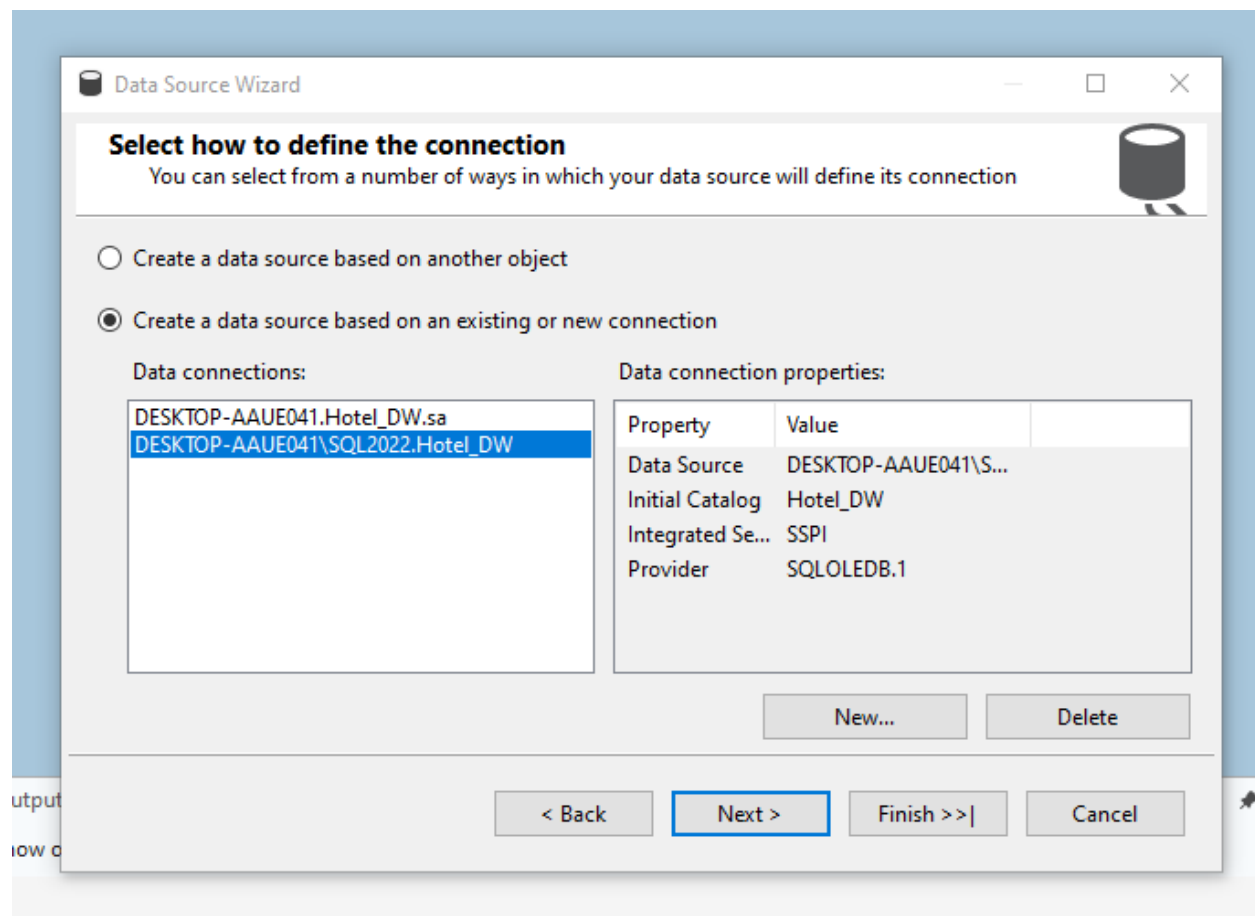
Overview

A multidimensional OLAP cube called "Cube_Hotel_DW" was created using SQL Server Analysis Services (SSAS) in Visual Studio. The cube pre-calculates data from the Hotel_DW warehouse, enabling fast reporting and analysis.

2.1 Data Source Definition

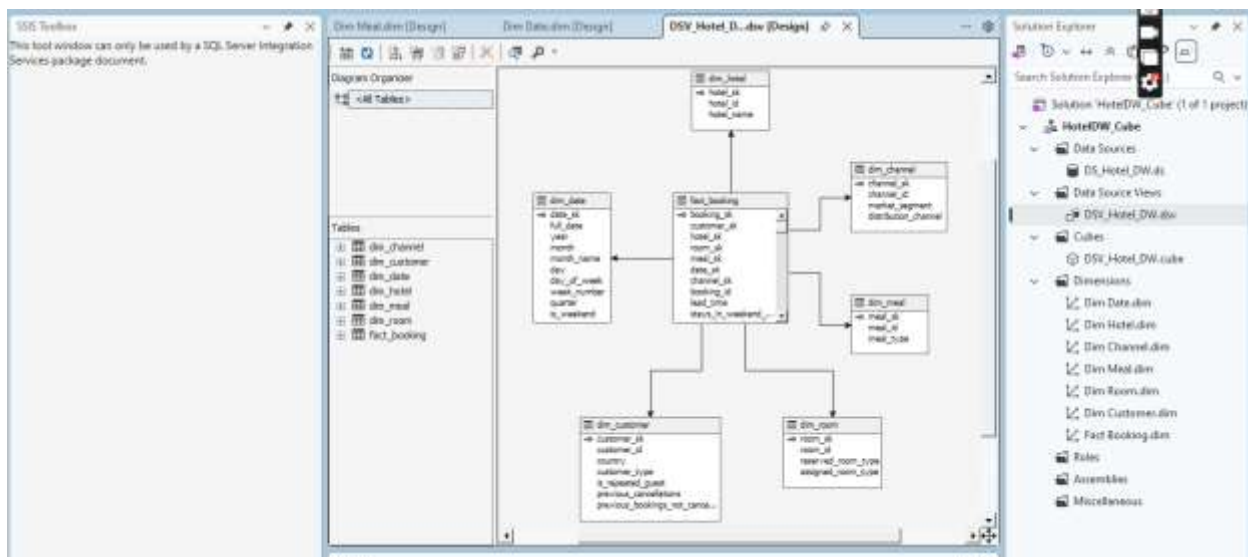
The first step in creating the SSAS project involved establishing a secure connection to the relational database that hosts the Hotel_DW data warehouse. A data source object was defined using the SQL Server Native Client provider, specifying the server name as DESKTOP-AAUE041\SQL2022 and the database name as Hotel_DW. This data source serves as the bridge between the Analysis Services instance and the relational data, allowing the cube to read data directly from the source tables during processing.

- Server - DESKTOP-AAUE041\SQL2022
- Data Base - Hotel_DW



2.2 Creating the Data Source View (DSV)

A Data Source View was created as a logical abstraction layer over the physical database tables. This DSV includes only the tables required for analysis: fact_booking as the measure group table, and all six dimension tables including dim_customer, dim_date, dim_hotel, dim_meal, dim_room, and dim_channel. Within the DSV designer, the star schema relationships were visually established by dragging foreign key columns to corresponding primary key columns. The relationships were verified to ensure proper joins between fact_booking.date_sk and dim_date.date_sk, fact_booking.hotel_sk and dim_hotel.hotel_sk, and similarly for all other dimensions.



2.3 Cube Creation and Measure Groups

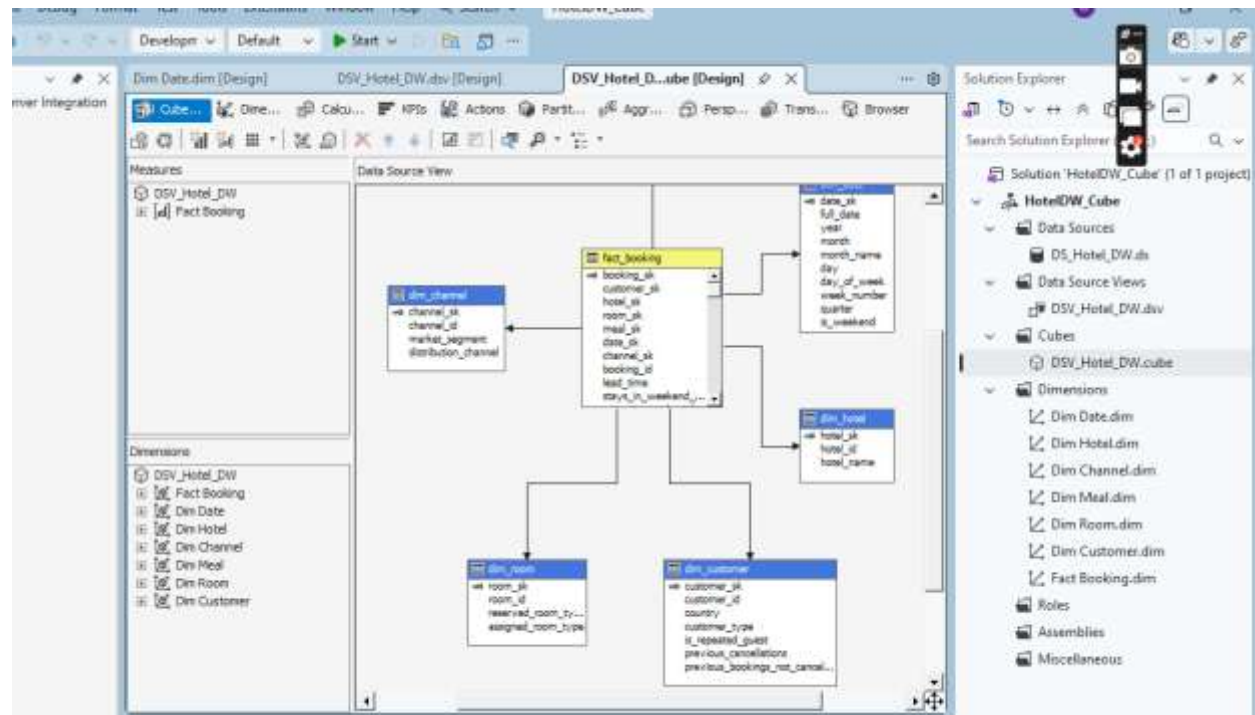
The cube includes a comprehensive set of measures selected from the fact_booking table to support hotel performance analysis.

Revenue serves as the primary financial measure for total income analysis. ADR (Average Daily Rate) provides insight into pricing strategies and hotel competitiveness. Lead Time reveals booking patterns and customer planning behavior. Total Nights indicates length of stay trends essential for occupancy planning.

Adults and Children counts enable family travel pattern analysis, while Total Guests captures complete party size information. Required Parking Spaces helps analyze parking demand patterns, and Special Requests tracks guest service requirements.

Booking Changes measures booking flexibility and operational efficiency, while Days in Waiting List indicates demand pressure and waitlist effectiveness.

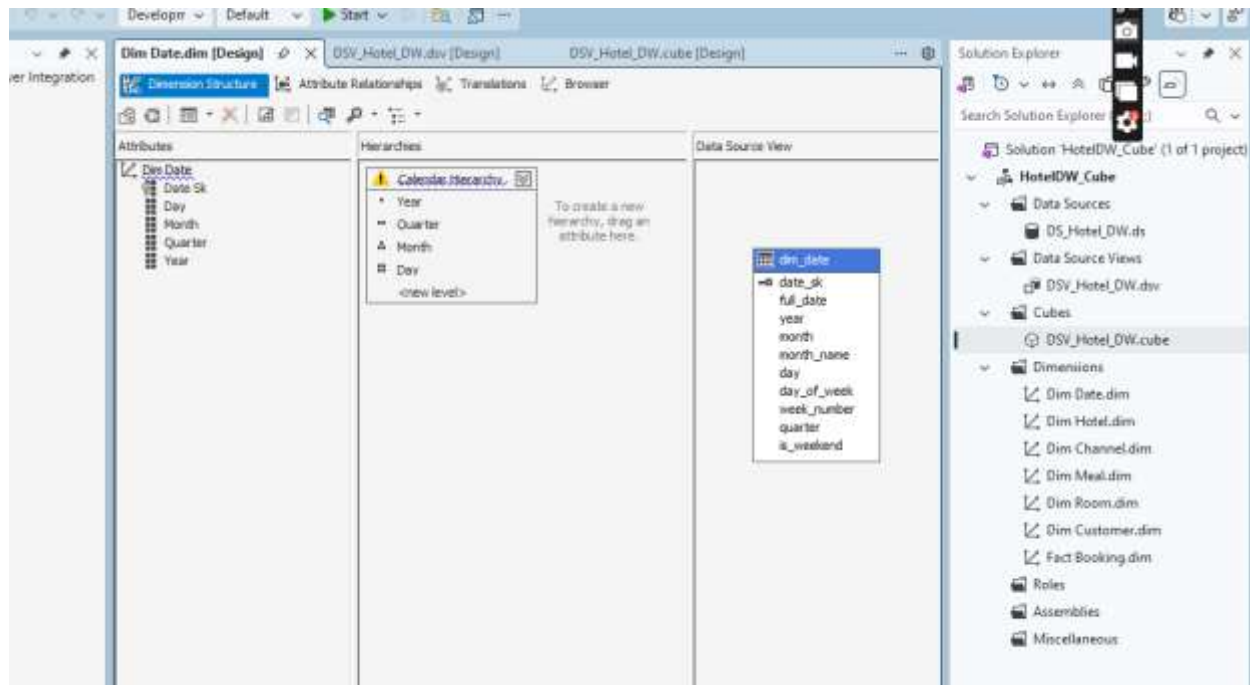
All measures were configured with appropriate aggregations, primarily using SUM for additive measures and AVERAGE for ratio measures like ADR.



2.4 Dimension Design and Hierarchies

Dimension Hierarchies

- Calendar Hierarchy in dim_date follows Year → Quarter → Month → Day, allowing progressive drill-down from annual summaries to daily details for identifying seasonal patterns and high-performing periods.
- Customer Hierarchy in dim_customer follows Customer Type → Country, enabling analysis of booking patterns across different customer segments and geographic origins.
- Hotel Hierarchy includes Hotel Name as a single-level hierarchy, enabling direct comparison between Resort Hotel and City Hotel.
- These hierarchies provide the foundation for drill-down and roll-up operations in OLAP analysis.



2.5 Dimension Usage and Dimensionality Mapping

The Dimension Usage tab was configured to establish how each dimension relates to the fact_booking measure group. This setup ensures that the OLAP cube can properly break down numeric measures across different analytical perspectives, enabling operations such as slice, dice, and drill-down. The Dimension Usage tab is critical because it defines how the cube understands relationships between the fact table and dimensions, allowing users to analyze data from multiple angles.

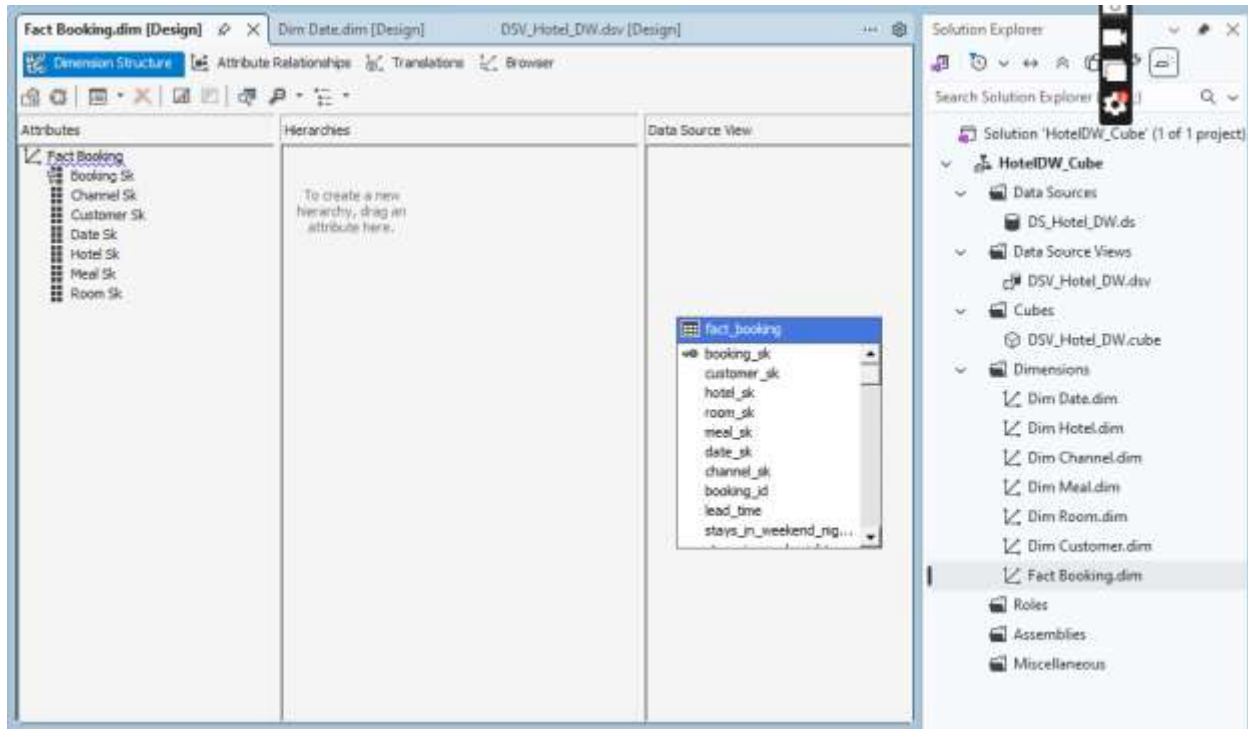
Relationship Type: The fact_booking table is connected to all six dimensions including dim_date, dim_hotel, dim_customer, dim_room, dim_meal, and dim_channel using Regular relationships. This follows the star schema approach where all dimensions are directly related to the central fact table, enabling efficient query performance and straightforward data navigation.

Key Linking: Primary keys from dimension tables were mapped to corresponding foreign keys in the fact_booking table. For example, dim_date.date_sk was linked to fact_booking.date_sk, dim_hotel.hotel_sk to fact_booking.hotel_sk, and similarly for dim_customer.customer_sk, dim_room.room_sk, dim_meal.meal_sk, and dim_channel.channel_sk. These mappings ensure proper join conditions during query processing.

Granularity: This mapping sets the cube's granularity to the individual transaction level. By linking each dimension to its corresponding key level, every booking record in fact_booking can be analyzed independently. The granularity is determined by connecting each dimension to the fact

table at the key level, ensuring that the cube maintains detail at the booking level while supporting aggregations at higher levels.

Analytical Impact: This configuration is essential for maintaining data accuracy and correctness. When users apply filters to dimensions such as selecting a specific hotel like "Resort Hotel" or a particular year like "2015", the SSAS engine automatically aggregates the relevant measures such as Revenue and ADR without needing to duplicate or copy data. The engine efficiently handles these calculations across dimensions, providing fast and accurate results for business analysis.

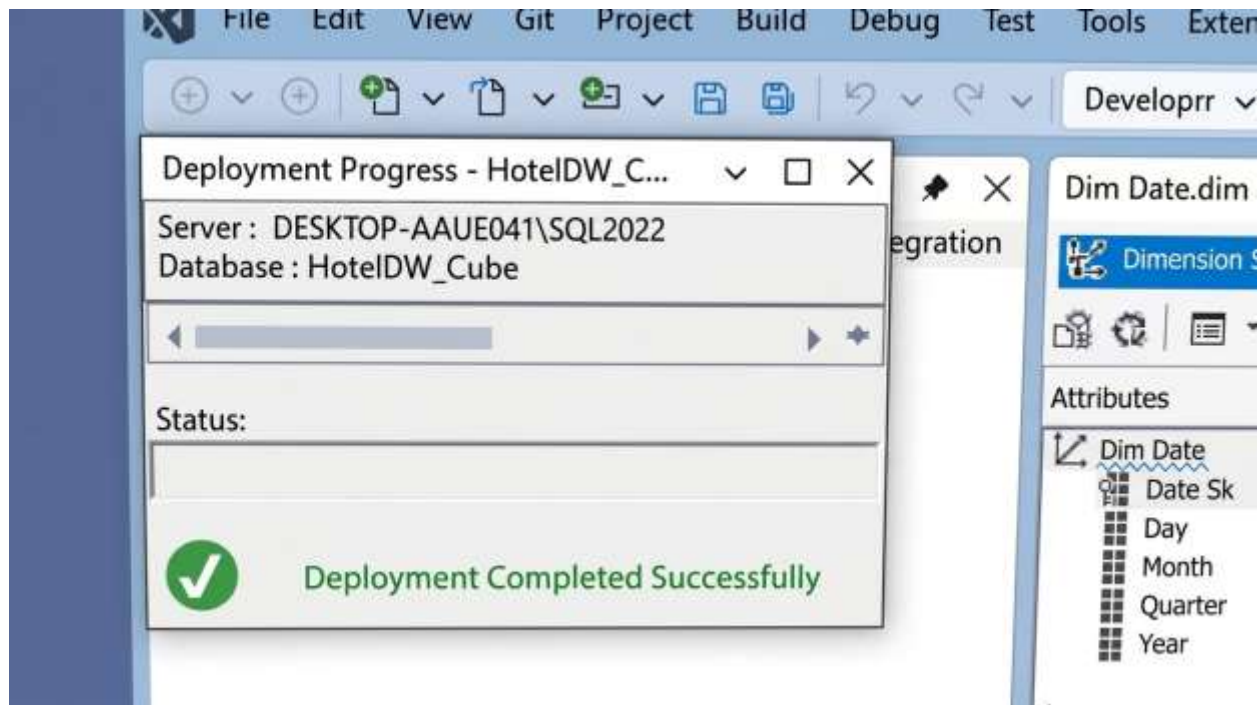


2.6 Cube Deployment and Processing

With the logical design now complete, the cube project was deployed to the Analysis Services instance on my machine DESKTOP-AAUE041\SQL2022. The deployment process involved creating the cube database on the server and processing all dimensions and measures according to the defined hierarchies and relationships.

The next step was processing the cube. During this phase, the SSAS engine reads the relational data from the Hotel_DW data warehouse, performs the pre-calculated aggregations defined by the hierarchies, and builds the multidimensional structure that enables fast query responses. All dimensions including dim_date, dim_hotel, dim_customer, dim_room, dim_meal, and dim_channel were processed successfully. The fact_booking measure group was processed without any errors, and all hierarchies including the Calendar Hierarchy (Year → Quarter → Month → Day) were built correctly.

The cube deployment and processing completed without any problems. There were no errors or warnings during the process, and the cube was successfully deployed and available for client connections from Excel and Power BI for OLAP analysis.



Step 3: Demonstration of OLAP operations

To demonstrate the analytical capabilities of the Cube_Hotel_DW SSAS cube, Microsoft Excel was used as the client tool. Excel was connected to the Analysis Services database where the cube was deployed. Using Excel PivotTables connected to the SSAS cube, the star schema structure of Hotel_DW was explored. The five fundamental OLAP operations—Slice, Dice, Drill-Down, Roll-Up, and Pivot—were performed to showcase how business users can interact with and analyze data warehouse metrics.

3.1 Slice Operation

Theoretical Description: The Slice operation filters a multidimensional cube by selecting a single dimension value, creating a two-dimensional subset of data. This operation simplifies analysis by focusing on one specific segment of the cube, allowing users to examine performance metrics for a particular business entity without distraction from other data.

Practical Implementation:

Action: The Calendar Hierarchy from Dim Date was placed in the Rows area of the PivotTable, and the Revenue measure from Fact Booking was placed in the Values area to display total revenue across time periods. The Hotel Name dimension from Dim Hotel was then added to the Filters area.

Filtering: A filter was applied to the Hotel Name dimension, selecting only "Resort Hotel". This eliminated all data related to City Hotel from the analysis.

Analytical Result: The resulting PivotTable displayed revenue figures exclusively for Resort Hotel across all years. This effectively sliced through the multidimensional cube to remove City Hotel data, enabling focused analysis on a single property's performance. The operation demonstrates how business users can isolate specific business segments for detailed examination.

	E	F	G	H
	Hotel Name	City Hotel		
	Row Labels	Revenue	Booking Changes	
	2015	3291706.26	1933	
	2016	11789567.5	7323	
	2017	10198168.42	5608	
	Grand Total	25279442.18	14864	

3.2 Dice Operation

Theoretical Description: Unlike Slice which filters on a single dimension, the Dice operation applies multiple filters simultaneously across two or more dimensions. This creates a sub-cube—a smaller, more focused subset of data that represents the intersection of multiple selection criteria. Dice enables analysis of highly specific scenarios where multiple conditions must be met.

Practical Implementation:

Action: Building upon the previous Slice operation, the filter for Resort Hotel was maintained. An additional filter was then applied to the Meal Type dimension from Dim Meal.

Filtering: The Meal Type filter was set to "HB" (Half Board), selecting only bookings where guests chose the half-board meal plan. The existing Resort Hotel filter remained active.

Analytical Result: The PivotTable now displayed a highly targeted sub-cube showing revenue for Resort Hotel bookings with HB meal plans. This dice operation allowed analysis of a specific combination—understanding how the half-board meal plan performs at the Resort Hotel. The operation demonstrates how analysts can narrow their focus to very specific business scenarios by applying multiple simultaneous filters.

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3.3 Drill-Down Operation

Theoretical Description: The Drill-Down operation navigates from higher-level summaries down to more detailed levels of data by expanding hierarchies. This operation allows users to start with broad overviews and progressively explore underlying details to identify trends, patterns, or anomalies that exist at lower granularities.

Practical Implementation:

Action: The Calendar Hierarchy from Dim Date was placed in the Rows area of the PivotTable, initially showing revenue summarized at the Year level. The Revenue measure was placed in Values.

Execution: The expand (+) icon next to "2015" was clicked, revealing the four quarters within that year. Quarter 3 was then expanded further, revealing the individual months of July, August, and September. As each level was expanded, the OLAP engine automatically recalculated the revenue figures at the new granularity.

Analytical Result: Starting from annual revenue of \$6.8 million for 2015, drilling down revealed that Quarter 3 contributed the largest portion, and further drilling showed that July had the highest

revenue among the three months. This operation demonstrates how users can navigate from high-level summaries down to transaction-level detail to identify specific time periods driving overall performance.

A11				1
	A	B	C	D
1				
2		#SPILL!		
3	Row Labels	Revenue	Booking Changes	
4	2015			
5	3			
6	7	1146860.64	406	
7	8	1698554.4	716	
8	9	1666153.56	799	
9	4	2306541.71	1987	
10	2016			
11	1			
12	1	403694.34	557	
13	2	788224.44	677	
14	3	1226925.78	1074	
15	2	5077203.66	3321	
16	3	7271184.03	3567	
17	4	4103348.14	3554	
18	2017			
19	1	3312418.75	2882	
20	2	7240764.85	3846	
21	3	6481578.4	3014	
22	Grand Total	42723452.7	26400	
	< >	Sheet2	Sheet3	Sheet1 +

3.4 Roll-Up Operation

Theoretical Description: The Roll-Up operation is the inverse of Drill-Down. It collapses hierarchical levels to provide higher-level summaries, reducing data granularity. This operation allows users to zoom out from detailed views to understand broader trends and patterns across larger categories or time periods.

Practical Implementation:

Action: Starting from the month-level view achieved in the Drill-Down operation, where individual months were visible within Quarter 3 of 2015, the Collapse operation was performed.

Execution: The collapse (-) icon next to Quarter 3 was clicked, which removed the month-level detail and returned to quarterly aggregation. Quarter 3 was then collapsed into the year level, showing revenue summarized at annual granularity. As each level was collapsed, the OLAP engine aggregated the detailed data to provide higher-level summary values.

Analytical Result: The detailed variations for individual months were removed, and the revenue figures were rolled up to show quarterly and then annual totals. This operation demonstrates how users can progressively broaden their view to understand overall trends after examining specific details, enabling both micro and macro perspectives of the same data.

A7 : X ✓ fx 2016				
	A	B	C	D
1				
2		#SPILL!		
3	Row Labels	Revenue	Booking Changes	
4	2015			
5	3	4511568.6	1921	
6	4	2306541.71	1987	
7	2016	18870580.39	12750	
8	2017			
9	1	3312418.75	2882	
10	2	7240764.85	3846	
11	3	6481578.4	3014	
12	Grand Total	42723452.7	26400	
13				
14				

3.5 Pivot Operation

Theoretical Description: The Pivot operation, also known as rotation, changes the orientation of data axes in a multidimensional cube. This operation does not alter the underlying data or calculations but presents the same information from a different perspective, allowing users to view relationships from alternative angles.

Practical Implementation:

Action: The dimensions were reorganized within the PivotTable Fields pane. The Calendar Hierarchy was moved from the Rows area to the Columns area, changing its orientation from vertical to horizontal. Simultaneously, the Hotel Name dimension was moved from the Filters area to the Rows area.

Execution: The Calendar Hierarchy now appeared across the top of the PivotTable horizontally, displaying years and quarters as column headers. Hotel Name appeared down the left side vertically as row headers. The Revenue measure remained in the Values area.

Analytical Result: The entire report layout was transformed. Time series data was now displayed across columns, making year-over-year comparison easier. Hotel data was displayed down rows, allowing side-by-side comparison between Resort Hotel and City Hotel. This pivot operation demonstrated how the same underlying data can be presented in different orientations to support different analytical questions and visual preferences.



The screenshot shows an Excel PivotTable with the following structure:

Column Labels		2016			2017			Total Revenue		Total Booking Changes	
Revenue	Booking Changes	Adults	Revenue	Booking Changes	Adults	Revenue	Booking Changes	Adults	Revenue	Booking Changes	Adults
6018110.31	3908	40262	18870500.39	12799	154964	17034762	9743	76418	42723402.7	26400	

Step 4: Power BI Reports

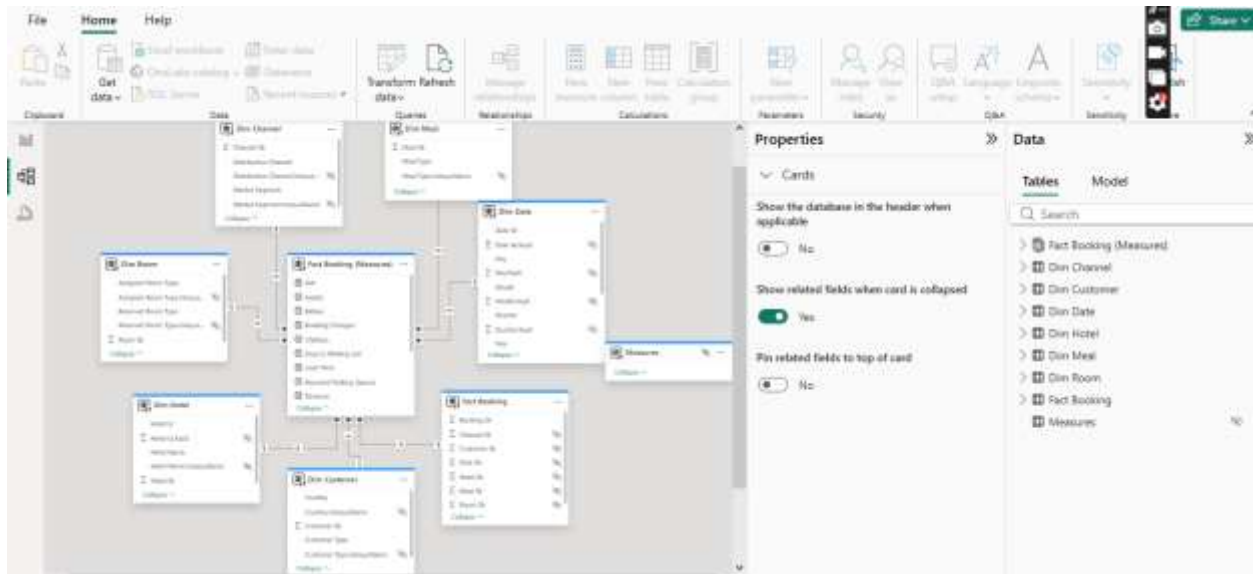
Overview and Architectural Approach

To fulfill the final business intelligence requirement of this assignment, Microsoft Power BI Desktop was utilized to design four interactive and insightful reports. Unlike traditional approaches that import flat files, a Live Connection was established directly to the deployed SSAS Multidimensional Cube named Cube_Hotel_DW. This architectural decision was made to ensure a "single version of truth" across all reports, leveraging the pre-calculated OLAP aggregations and the rich hierarchies defined in Step 2.

The Live Connection approach offers several advantages. It eliminates data duplication by keeping all data within the SSAS cube, ensures consistent calculations across all reports, and provides optimal query performance through pre-aggregated measures. The existing star schema relationships and the Calendar Hierarchy (Year → Quarter → Month → Day) created in the cube are fully utilized in Power BI, enabling seamless drill-down and roll-up operations.

All four reports were designed with a consistent visual theme, including professional color schemes, borders, shadows, and clear titles. The reports were published to Power BI Service for

cloud-based accessibility, allowing stakeholders to interact with the reports from any device. The following sections detail each report's purpose, implementation approach, and analytical impact.

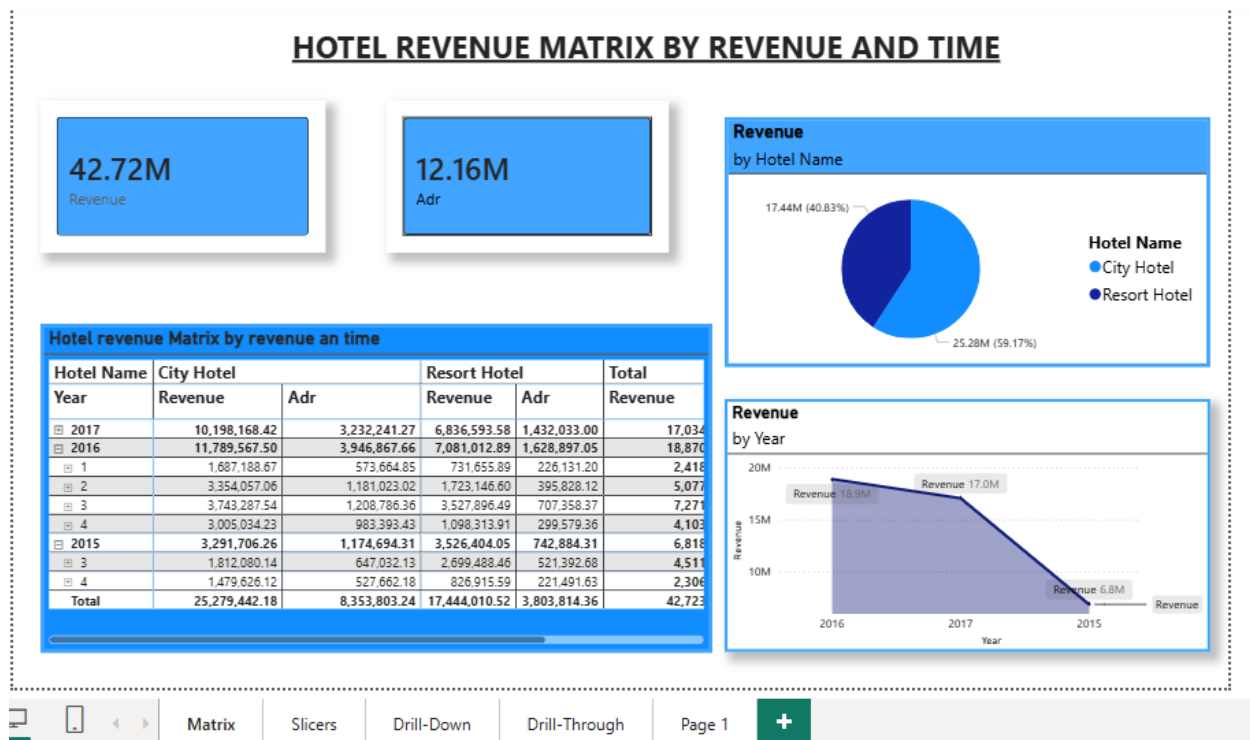


4.1 Report 1: Hotel Revenue Matrix by Year and Hotel

Description: This report uses a Matrix visual to display hotel revenue performance across time periods and hotel properties, allowing users to view data at both summary and detailed levels through expandable hierarchies.

Implementation Details: The Calendar Hierarchy (Year → Quarter → Month) was placed in Rows to enable vertical drill-down, while Hotel Name was placed in Columns for side-by-side comparison between Resort Hotel and City Hotel. Revenue and ADR measures were added to Values, providing both total income and rate analysis. KPI cards above the matrix show Total Revenue (\$42.7M), Average ADR (\$101.83). Professional formatting includes borders, shadows, and a consistent blue and green color scheme.

Analytical Impact: Users can start with annual summaries, expand to quarterly and monthly details, and compare performance between hotels. The report reveals that City Hotel generates higher total revenue while both hotels maintain similar ADR patterns, helping users understand whether revenue differences stem from occupancy or pricing strategies.



4.2 Report 2: Cascading Slicers Dashboard

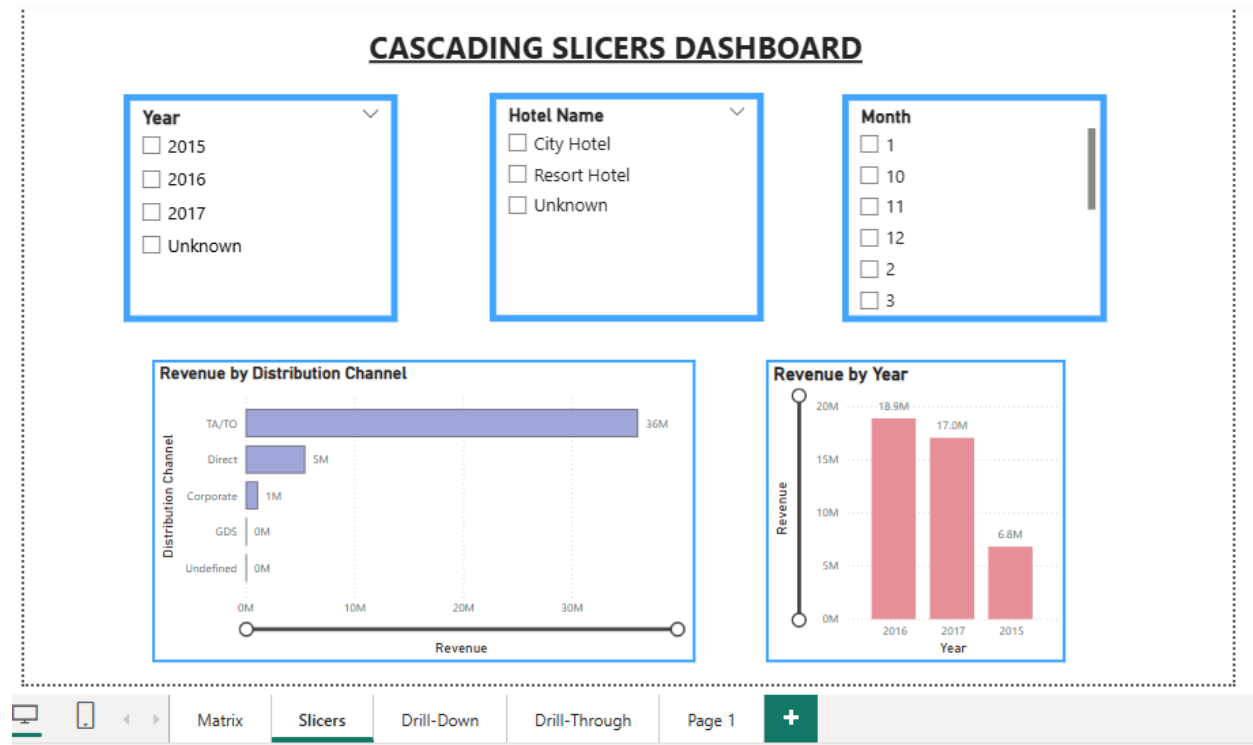
Description: This report features an interactive dashboard with cascading slicers that dynamically filter each other based on user selections. When a value is selected in one slicer, the options available in subsequent slicers automatically update to show only valid combinations, preventing users from selecting invalid filter combinations and improving the overall analytical experience.

Implementation Details: Three cascading slicers were added to the canvas: Year slicer from Dim Date, Month slicer from Dim Date, and Hotel Name slicer from Dim Hotel. The cascading behavior is achieved through Power BI's native cross-filtering functionality, where selecting a value in the Year slicer automatically filters the Month slicer to display only months within that year. This ensures users cannot select a month that does not exist in the chosen year.

To provide visual feedback, a Stacked Bar Chart was added showing Revenue by Market Segment, and a Line Chart displaying Revenue by Year. These visuals update instantly when slicer selections change, allowing users to see the impact of their filter choices immediately. A Card visual displays Total Revenue, which also responds dynamically to all slicer selections.

User Experience: When a user selects "Resort Hotel" in the Hotel slicer, all charts automatically update to show only Resort Hotel data. If a user then selects "2016" in the Year slicer, the Month slicer updates to show only months from 2016, and all charts reflect the combined filter of Resort Hotel in 2016. This cascading logic prevents empty results and guides users toward valid analytical paths.

Analytical Impact: The cascading slicer design enables users to explore data progressively, starting with broad filters and narrowing down to specific combinations. The immediate visual feedback helps users understand how each filter affects the data, supporting exploratory analysis and hypothesis testing. The combination of slicers and dynamic charts provides a comprehensive tool for understanding revenue patterns across hotels, time periods, and booking channels.

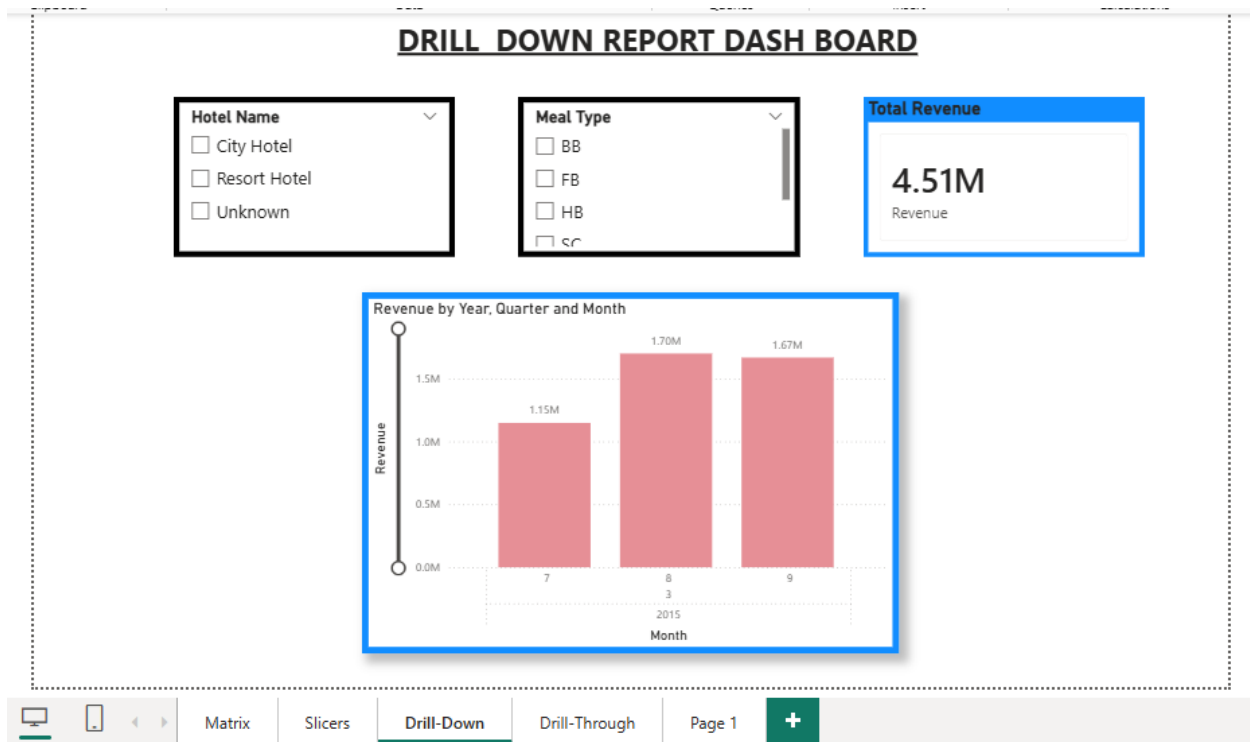


4.3 Report 3: Hierarchical Revenue Trend Analysis (Drill-Down)

Description: This report enables time series analysis through drill-down functionality, allowing users to explore revenue trends from annual summaries down to monthly details. The interactive chart transforms static data into an exploratory analytical tool.

Implementation Details: A Stacked Column Chart was created with Calendar Hierarchy (Year → Quarter → Month) on the X-axis and Revenue on the Y-axis. Drill-down mode was enabled, allowing users to double-click on any year to view quarters, and double-click on quarters to view months. Hotel Name and Meal Type slicers were added for cross-dimensional filtering, and a Card visual displays Total Revenue that updates dynamically.

Analytical Impact: Users can start with a macro view of annual revenue, drill down to identify which quarters contributed most to performance, and further drill down to pinpoint specific high-performing months. This hierarchical navigation helps identify seasonal patterns and supports detailed trend analysis without switching between multiple reports.



4.4 Report 4: Drill-Through Navigator

Description: This report implements the "overview first, details on demand" paradigm using a two-page drill-through design. Users can start with summary visualizations and seamlessly navigate to detailed transactional pages for deeper investigation, enabling efficient exploratory analysis without cluttering the main dashboard.

Implementation Details: The source page (Drill-Down Report) contains a column chart showing Revenue by Date with drill-down capabilities. The target page (Drill-Through Details) contains a detailed Table visual displaying booking information including Hotel Name, Year, Month, Revenue, ADR, Meal Type, Market Segment, and Cancellation Status. KPI cards at the top show Total Revenue and Total Bookings for context.

The drill-through configuration was established by dragging Year, Month, and Hotel Name fields into the "Add drill-through fields here" area on the target page. When users right-click on any bar in the source page and select Drill-Through, the filter context is automatically passed to the target page, showing only bookings matching the selected time period and hotel.



BOOKING DETAILS - DRILL THROUGH

Revenue

6.82M

Revenue

Average

1.92M

Adr

Drill Through Table

Hotel Name	Year	Month	Revenue	Adr	Meal Type	Market Segment	Total Nights
City Hotel	2015	10	0.00	0.00	BB	Complementary	26
City Hotel	2015	10	20,960.50	9,242.50	BB	Corporate	238
City Hotel	2015	10	56,580.42	21,016.45	BB	Direct	478
City Hotel	2015	10	175,623.88	65,758.66	BB	Groups	2247
City Hotel	2015	10	179,201.28	72,158.47	BB	Offline TA/TO	2291
City Hotel	2015	10	239,320.05	76,432.70	BB	Online TA	2250
City Hotel	2015	10	1,558.97	290.71	HB	Direct	10
City Hotel	2015	10	19,619.20	8,582.60	HB	Groups	202
City Hotel	2015	10	93,953.77	42,870.38	HB	Offline TA/TO	929
City Hotel	2015	10	578.88	199.17	HB	Online TA	6
City Hotel	2015	10	80.00	80.00	SC	Corporate	1
Total			6,818,110.31	1,917,578.62			74588



Matrix

Slicers

Drill-Down

Drill-Through

